

STREAMING DATAPATH

Shared SDRAM
left/right images

Burst reader

Input FIFOs

Line / history buffers
raster reuse and
right-image disparity taps

P
absolute-difference lanes
$$A_d(x, y) = |L(x, y) - R(x - d, y)|$$

Window SAD accumulators
horizontal +
vertical box sums

Winner-take-all
best cost, disparity,
optional second-best

Output FIFO

Burst writer

Shared SDRAM
best state /
disparity map

Performance counters
total, active, read stalls,
write stalls, pixels

Frame / disparity-group controller
coordinates, strides, border validity,
lane scheduling, start/done

Baseline multi-pass schedule: each raster pass evaluates P disparities. The best-cost/disparity map in SDRAM carries state between disparity groups. Increasing P reduces passes but raises lane and buffer resources.